# Work Packet · WO-DEMO-0001 · CAT Pumps 5CP6120 PROPOSED

Demo Plant 1 (fictional) · asset FOG-PUMP-03 · generated 2026-10-30 · packet draft-0.1

**DRAFT — reviewed by a technician only when marked; not approved for use until the supervisor approval below is recorded.**

**Labels.** VERIFIED quoted or computed from the cited manufacturer document page, or from a deterministic check; verified against the DOCUMENT, not yet verified in the field PROPOSED drafted from the documents for technician review; must be confirmed or corrected before use UNKNOWN not stated by any cited source; must be supplied by the technician or site; never guessed

**Scope** PROPOSED

Reseal (seal kit) and valve service (two valve kits); plungers inspected, replaced only if scored symptom 'weeping at manifold, pressure drop' + service manual p.3 'Generally Plungers do not need to be replaced'

**Seal material** PROPOSED · SET BY DEMO TECHNICIAN the data sheet lists five seal materials (NBR standard, FPM .0110, EPDM .0220, HT .3000, STHT .3400); the pump's suffix decides which kit numbers apply (data sheet p.4)

## Sources

| Id | Document | Revision | Pages used | Hash |
|---|---|---|---|---|
| DS | CAT Pumps 5CP6120 / 5CP6120CS / 5CP6120CSSG1 Data Sheet | Rev G 5/22 | 1, 2, 3, 4 | a3b9fdfeeb90bc32 |
| SM | CP Plunger Pump Service Manual (3CP, 5CP, 7CP) | unknown (text shows 'Rev J' marker; confirm on the PDF cover) | 1, 2, 3, 5, 6 | f05fb18880eadb32 |
| FIELD | Rev F 11/20 copy photographed on site 2026-10-30 | unknown | | — |

## Safety

**Isolation — site procedure required** UNKNOWN

Isolate per the SITE-APPROVED procedure and verify zero pressure before any step below. This packet does not generate lockout or isolation steps; a qualified person applies the site procedure and signs below.

## Manufacturer warnings

- VERIFIED *All systems require both a primary pressure regulating device (i.e., regulator, unloader) and a secondary pressure safety relief device (i.e., pop-off valve, safety valve). Failure to install such relief devices could result in personal injury or damage to the pump or to system components.* (SM p.1)
- VERIFIED *Read all CAUTIONS and WARNINGS before commencing service or operation of any high-pressure system.* (DS p.4)
- VERIFIED *DO NOT RUN PUMP WITHOUT OIL IN CRANKCASE.* (SM p.1)

## Hazards to review in the site JHA PROPOSED

- stored pressure in the manifold and hoses
- hot surfaces after operation
- EPDM elastomers require silicone-base lubricant (SM p.5) — chemical compatibility of lubricant

site JHA decides controls and PPE

## Qualifications UNKNOWN

site-defined; not stated by the manufacturer documents

## Parts VERIFIED

DS p.3 SERVICE PARTS and p.4 SEAL KITS / VALVE KITS; kit coverage checked by tools/parts-breakdown (11/12 pass, 1 not_covered — item 120 NBR)

| Part | Description | Qty | Basis | Condition | |
|---|---|---|---|---|---|
| 33628 | Kit, Seal – Standard NBR | 1 | DS p.3; 'One (1) seal kit required per pump' DS p.4 | ONLY if the pump's seal material is NBR (standard). Other materials: FPM 31628, EPDM 31648, HT 31638, STHT 76536 (DS p.4) | PROPOSED · CONFIRM BY DEMO TECHNICIAN |
| 30820 | Kit, Valve – Standard NBR | 2 | DS p.3; 'Two (2) valve kits required per | NBR standard; FPM 31263, EPDM 31259 (DS p.4) | PROPOSED |

| | | | | | |
|---|---|---|---|---|---|
| | | | pump' DS p.4; SM p.2 'Two Valve Kits are needed for complete valve change' | | |
| 6107 | Oil, Bottle (21 oz) ISO 68 Hydraulic | 1 | DS p.3 service parts; crankcase capacity 17 oz / 0.50 l (DS p.1, SM p.1) | — | VERIFIED |
| ~~6124~~ | ~~Gasket, Liquid (3 oz)~~ | ~~1~~ | ~~DS p.3; SM p.5 'For certain applications apply liquid gasket... See Tech Bulletin 053'~~ | ~~only if Tech Bulletin 053 applies to this model/application~~ | PROPOSED · STRIKE BY DEMO TECHNICIAN |

**Kit coverage check** VERIFIED

seal kit items 98, 106, 121, 125 and valve kit items 162, 163, 164, 166, 167, 168, 172 at their expected per-pump quantity on the parts list; item 120 is NOT verified for the standard NBR (STD) seal kit — DS p.4 shows no NBR (STD) part number for item 120, only STHT (.3400) (49436); ordering the standard NBR kit alone will not supply a replacement for item 120 (DS p.2 vs p.4)

| Item | Part | Inspect, replace if worn | Qty | Source | |
|---|---|---|---|---|---|
| 90 | 46841 | Plunger (M20 x 50), ceramic | 3 | DS p.2; SM p.3 'Examine Ceramic Plunger for cracks or scoring' | PROPOSED |
| 100 | 46749 | Retainer, Seal | 3 | DS p.2; SM p.3 'Remove the Seal Retainer ... examine for wear' | PROPOSED |
| 70 | 46838 | Seal, Oil, Crankcase | 3 | DS p.2; SM p.5 crankcase section | PROPOSED |

## Tools PROPOSED

| Tool | For | Source |
|---|---|---|
| M24 hex socket (CAT tool P/N 44046) | valve plugs, 5CP M22 thread | SM p.6 torque chart |
| M6 Allen key (CAT tool P/N 30941) | manifold head socket-head screws M8 (item 188) | SM p.6; DS p.2 |
| M10 hex (CAT tool P/N 25082) | plunger retainer M6 (item 99) | SM p.6 |
| Reverse pliers (CAT P/N 30696) | seats, seal cases, lo-pressure seals | DS p.3 optional parts; SM p.2 and p.5 |

| | | |
|---|---|---|
| Needle-nose pliers; two flat screwdrivers; soft mallet | valve assemblies; separating the manifold head | SM p.2 and p.3 |
| Torque wrench, in-lb range to 870 in-lb (72.5 ft-lb) | valve plugs 870 in-lb; manifold screws 115 in-lb; plunger retainer 55 in-lb | SM p.6 |
| Silicone-base lubricant | ONLY if EPDM elastomers | SM p.5 note |

## Draft work steps PROPOSED

Draft order from the service manual; a technician confirms, reorders or strikes steps. Step 1 is site-owned and is never generated.

1. Isolate and relieve pressure per the SITE procedure; verify zero pressure at the manifold. UNKNOWN
   site procedure (not in the manufacturer documents)
2. Remove the hex valve plugs (top discharge, bottom inlet). Examine the O-ring under each plug; replace if worn; lubricate new O-rings. PROPOSED
   SM p.2 §1–2
3. Remove each spring retainer by its tab; remove spring and valve; with reverse pliers remove the seat; remove the O-ring at the bottom of the chamber. Note: the 5CP6120 has an O-ring AND a backup ring on each seat. PROPOSED
   SM p.2 §3
4. Examine valve parts for pitting, gouges or wear; replace with the preassembled valve assemblies from the kits (two valve kits for a complete change). Immerse each new assembly in oil and push in until fully seated. PROPOSED
   SM p.2 §4–5
5. Remove the socket-head screws from the manifold head; rotate the crankshaft by hand to start separation; separate with two flat screwdrivers and light mallet taps, supporting the head. CAUTION: keep the manifold aligned with the ceramic plungers. PROPOSED
   SM p.3 disassembly §1–3 and CAUTION
6. Remove the seal retainer from each plunger rod and examine for wear; examine each ceramic plunger for cracks or scoring (replace only if damaged). PROPOSED
   SM p.3 §4–5
7. With the crankcase side of the manifold facing up: remove the lo-pressure seal, the press-in seal case, its O-ring, and the hi-pressure seal (usually by hand). PROPOSED
   SM p.5 disassembly §1–4
8. Reassemble seals: lubricate each seal chamber; press the hi-pressure seal in grooved side down (metal back facing out); fit new seal-case O-rings; press each seal case in until seated.

Service with the SAME seal type the pump was built with. PROPOSED

SM p.5 reassembly §1–4 and NOTE

9. Install seal retainers (tab down, holes toward the manifold); rotate the crankshaft so the two outside plungers extend equally; lightly lubricate the plungers; slide the manifold head on, supporting from underneath. PROPOSED

   SM p.3 reassembly §2–4

10. Fit the manifold socket-head screws and torque to 115 in-lb (9.58 ft-lb, 13 Nm). PROPOSED

    SM p.3 §5; torque chart SM p.6 'Manifold Screw M8'

11. Torque the valve plugs to 870 in-lb (72.5 ft-lb, 98 Nm). PROPOSED

    SM p.6 'Valve Plugs 3CP, 5CP M22'

12. Crankcase check: inspect oil seals and rear cover, drain plug and bubble gauge for leaks; check oil level and for water in oil; rotate the crankshaft by hand for smooth bearing movement. PROPOSED

    SM p.5 crankcase §1–5

13. Fill the crankcase to specification (17 oz / 0.50 l). Do not run the pump without oil. PROPOSED · CONFIRM BY DEMO TECHNICIAN

    DS p.1; SM p.1 LUBRICATION

14. Return to service per the site procedure; run and check for leaks at the manifold; confirm rated performance (7.4 gpm at 1725 rpm, 100–1500 psi). PROPOSED

    site procedure + DS p.1 specifications; acceptance limits are site-defined (unknown)

## Measurements and tolerances

| Measurement | Value | Source | |
|---|---|---|---|
| Crankcase oil fill | 17 oz (0.50 l) | DS p.1 'Crankcase Capacity 17 oz 0.50 l'; SM p.1 '5CP-17 oz.' | VERIFIED |
| Torque, manifold head screws (M8) | 115 in-lb / 9.58 ft-lb / 13 Nm | SM p.6 | VERIFIED |
| Torque, valve plugs (5CP, M22) | 870 in-lb / 72.5 ft-lb / 98 Nm | SM p.6 | VERIFIED |
| Torque, plunger retainer (M6) | 55 in-lb / 4.6 ft-lb / 6.2 Nm | SM p.6 | VERIFIED |
| Rated flow and pressure | 7.4 gpm (28 lpm); 100–1500 psi (6.9–103 bar); 1725 rpm | DS p.1 | VERIFIED |
| Max liquid temperature (NBR) | 160 °F (71 °C) | DS p.1 | VERIFIED |

| | | | |
|---|---|---|---|
| Functional-test acceptance (pressure held, leak-free minutes) | — (unknown) | site-defined | UNKNOWN |

## Exceptions and unknowns

- Seal material suffix not confirmed; kit numbers above assume NBR standard UNKNOWN → nameplate photo (task 10)
- Whether Tech Bulletin 053 (liquid gasket) applies to this model/application UNKNOWN → CAT Pumps tech bulletin 053
- Plunger and seal-retainer condition decides whether items 90/100 are needed UNKNOWN → inspection at steps 6
- Your printed data sheet is Rev F (11/20); current is Rev G (5/22). Kit numbers unchanged; three parts-list rows differ VERIFIED → use Rev G

## Review and sign-off

| Role | State | By | At |
|---|---|---|---|
| Isolation (qualified person) | required | | |
| Technician review | reviewed | Demo Technician | 2026-10-30T14:03:00Z |
| Supervisor approval | pending | | |

### Technician edits

- `set job.seal_material_suffix = NBR (standard) — nameplate shows no suffix — Demo Technician 2026-10-30T14:00:00Z · confirmed on the nameplate photo`
- `confirm parts.required[0] — Demo Technician 2026-10-30T14:01:00Z`
- `confirm steps.items[12] — Demo Technician 2026-10-30T14:02:00Z`
- `strike parts.required[3] — Demo Technician 2026-10-30T14:03:00Z · TB053 not applicable per distributor`

Technician signature / date

Supervisor signature / date

Isolation verified by / date

Labels: verified 14 · proposed 24 · unknown 9 · struck 1. A reviewed packet is still a draft until approved.